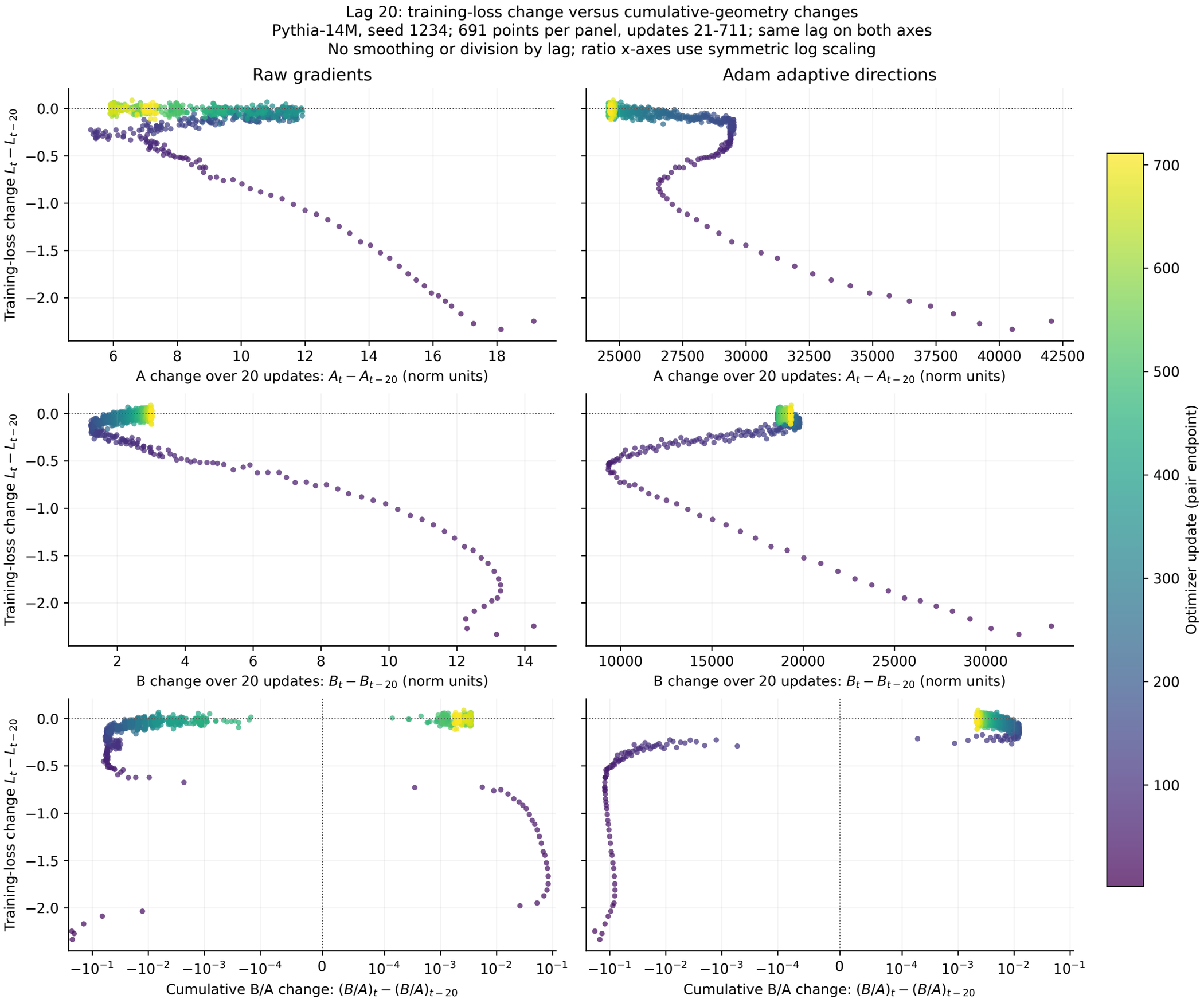

Lag 20: training-loss change versus cumulative-geometry changes
Pythia-14M, seed 1234; 691 points per panel, updates 21-711; same lag on both axes
No smoothing or division by lag; ratio x-axes use symmetric log scaling
Raw gradients
Adam adaptive directions
Training-loss change $L_t - L_{t-20}$
A change over 20 updates: $A_t - A_{t-20}$ (norm units)
B change over 20 updates: $B_t - B_{t-20}$ (norm units)
Cumulative B/A change: $(B/A)_t - (B/A)_{t-20}$
Optimizer update (pair endpoint)